METRIC
MIL-PRF-85045/33
18 May 2022

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, ROLLED FLEXIBLE RIBBON, TWELVE FIBER, SINGLE SUBUNIT, CABLE CONFIGURATION TYPE 4 (RIBBON CABLE), APPLICATIONS A AND B (AIRBORNE AND SHIPBOARD), CABLE CLASS SM AND MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

SCOPE:

This specification sheet addresses the requirements of a fiber optic cable containing a 12-fiber, rolled flexible ribbon with a 2-millimeter (mm) outer diameter. Coverage includes both fiber types and temperature ranges.

Design constraints. Figure 1 shows the cable construction and provides essential dimensioning detail for a fiber optic cable with a 2-mm outer diameter that contains a 12-fiber, rolled flexible ribbon. Cable construction consists of an outer jacket (with a 2-mm diameter), strength member (aramid yarn or aramid tape), water blocking material, and a rolled flexible ribbon (with 12 optical fibers). Following figure 1 are requirements for design and construction, material, and performance.

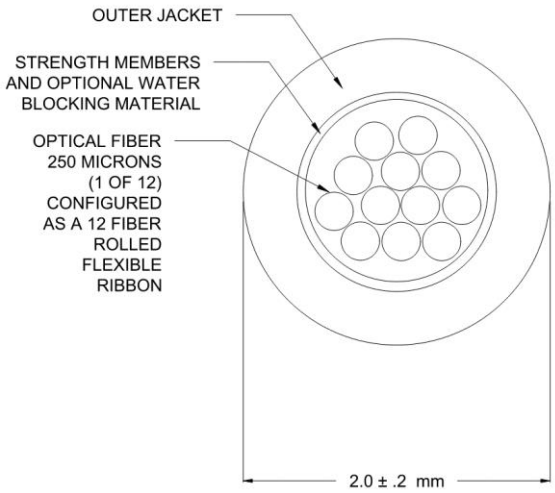


FIGURE 1. 12-fiber rolled flexible ribbon cable (single standalone subunit).

CLASSIFICATION:

Fiber optic cable configuration type. 4 (ribbon)

Fiber cable class: MM (graded-index, glass core and glass cladding, multimode).

SM (dispersion-unshifted, glass core and glass cladding, single mode).

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Part or identifying number (PIN):

12-fiber rolled flexible ribbon cable (multimode, temperature range 1, low smoke and zero halogen): M85045/33-01

12-fiber rolled flexible ribbon cable (single mode, temperature range 1, low smoke and zero halogen): M85045/33-02

12-fiber rolled flexible ribbon cable (multimode, temperature range 2): M85045/33-03

12-fiber rolled flexible ribbon cable (single mode, temperature range 2): M85045/33-04

DESIGN AND CONSTRUCTION:

Temperature ranges. Temperature range designations shall be in accordance with table I.

TABLE I. Temperature range designation.

Temperature range designation	Operating temperature		Non-operating temperature		Storage temperature	
1	-18 to 185 °F	-28 to 85 °C	-40 to 158 °F	-40 to 70 °C	-40 to 158 °F	-40 to 70 °C
2	-67 to 329 °F	-55 to 165 °C	-67 to 185 °F	-55 to 85 °C	-67 to 185 °F	-55 to 85 °C

Fiber: For temperature range 1 cable, MM fibers shall be MIL-PRF-49291/6 (PIN: M49291/6-03).

For temperature range 1 cable, SM fibers shall be MIL-PRF-49291/7 (PIN: M49291/7-01).

For temperature range 2 cable, MM fibers shall be MIL-PRF-49291/12 (PIN: M49291/12-02).

For temperature range 2 cable, SM fibers shall be MIL-PRF-49291/11 (PIN: M49291/11-02).

Buffer diameter. 250±15 micrometers (µm) after coloring (inking).

Ribbon fiber color coding. Each optical fiber in a ribbon shall be color coded for identification in accordance with TIA-598 (see 3.4.2.4.1 of MIL-PRF-85045). A visual inspection performed after the temperature life test shall show that the colorant on each optical fiber remains intact with no flaking or other degradation in addition to no change outside specified limits for appearance (color) on the three samples of rolled flexible ribbon.

Rolled flexible ribbon:

Dimensions and configuration:

Dimensions. See the requirement for the rolled flexible ribbon outer diameter.

Configuration. The rolled flexible ribbon shall be configured as an adhesive matrix (bonded layer) applied over 12 adjacent optical fibers that are positioned laterally (in a linear array) with the adhesive matrix applied intermittently. At any cross section along the length, only two adjacent fibers shall be joined by the adhesive matrix (see “DEFINITIONS”).

Fabrication. The rolled flexible ribbon shall have no crossovers, defective fibers, or splices.

Ribbon color coding. Not applicable for the ribbon adhesive matrix (see 3.4.2.4.2 of MIL-PRF-85045).

Tests for rolled flexible ribbon. Each test sample shall consist of 2 meters of finished cable or as specified in the test method. First, the specified testing shall be performed on the finished cable; then, the finished cable shall be deconstructed, and the group VII post-test examinations shall be performed on the rolled flexible ribbon lengths. Three samples shall be used for each test or as specified in the test method. Small end caps may be placed on the ends of the samples to eliminate end effects. A plastic cap or epoxy may be used. Test sample length and configuration for the ribbon twist and ribbon separation shall be as specified in those tests.

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Rolled flexible ribbon outer diameter. The rolled flexible ribbon shall be able to pass through a die with a 1.32±0.015-mm (0.052±0.0006-inch) diameter hole.

Ribbon strippability:

Method. One end of each rolled flexible ribbon shall be stripped to a minimum length of 30 mm with commercially available stripping tools to remove the binder (matrix) material and to remove the fiber coating (250-micron buffer). The test shall be performed both on unaged samples (from ribbon dimension test) and specimens of the rolled flexible ribbon from the finished cable that underwent temperature life followed by an inspection for ribbon delamination.

Requirement. No amount of fiber breakage in the rolled flexible ribbon shall occur, and any residue of coating that remains shall be readily removed with isopropyl alcohol wipes.

Ribbon twist (or twist robustness):

Method. Testing shall be performed on three samples of a 1-meter length of rolled flexible ribbon deconstructed from finished cable in accordance with TIA-455-141 for 20 cycles at a rate of 15±5 cycles per minute and at a rotation of 180±10 degrees in each direction for one cycle. The applied load shall have a weight of 500±25 grams. The cable length shall be sufficient to allow for a gauge length of 0.3±0.01 meter. A post-test visual inspection shall be performed.

Requirement. No amount of fibers shall become free and completely separate from the ribbon over the gauge length.

Ribbon delamination:

Method. Applicable except that three separate 2-meter samples of finished cable shall be included in the temperature life test versus performing a test under different conditions. After completion of the temperature test, the test samples of the rolled flexible ribbon (once outer cable jacket and aramid yarn are removed) shall be inspected for ribbon delamination.

Requirement. A visual inspection performed after the temperature life test shall show no delamination occurred on the three samples of rolled flexible ribbon.

Ribbon separation:

Method. The optical fibers in a minimum of 0.5-meter length of rolled flexible ribbon shall be separated either by hand using no tools or by hand using tools without the aid of applied heat.

Requirement. The minimum 0.5-meter length of optical fibers in a rolled flexible ribbon shall be separated with no damage, removal, discoloration, or chemical attack to the 250-micron buffer. Minor removal of buffer colorant is acceptable.

Finished cable:

Dimensions and configuration. See figure 1.

Mass per unit length. ≤5 kilograms (kg) per kilometer (km).

Outer cable jacket color:

Temperature range 1. Slate for class MM fiber. Yellow for class SM fiber.

Temperature range 2. The color of the cable jacket shall conform to the table for each optical fiber size as specified in requirement 1104 of MIL-STD-1678-1. The colors shall be within the light and dark color limits in accordance with 3.4.2.1.1 of MIL-PRF-85045.

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Material:

Cable jacket material. For temperature range 1, the cable jacket shall be composed of a low-halogen, low-smoke, low-toxicity polymer material.

Cable strain relief material requirement. The cable shall include a strength member (i.e., cable strain relief, such as one made from an aromatic polyimide, otherwise known as aramid yarn, or other solid form of a liquid crystal polymer, such as an aromatic polyester). For temperature range 2 only, cable strain relief material shall be braided. This cable strain relief material shall be capable of meeting the cable strain relief pullout force. Trade names for aramid yarns include Kevlar, Twaron, Nomex, New Star, and Teijinconex, or equal. Trade names for an aromatic polyester include Vectran, or equal.

Jacket red light transparency. Fiber breaks inside the cable jacket shall be easily recognized when a pulsed, red light from a laser (monochromatic light) is sent through the optical fiber in the cable. The monochromatic light shall have a nominal wavelength of 635 nanometers (nm), a nominal pulse rate of 3 hertz, and a minimum optical power level of 1 milliwatt (mW). A fracture in the optical fiber shall be created by placing the middle of a 1-meter length of fiber optic cable in a free form loop until a kink occurs in the cable jacket and the change in optical transmittance of the cable is reduced by at least 20 decibels (dB).

Number of fibers. 12 (12 fibers in one rolled flexible ribbon).

Concentricity. ≥0.80.

Short-term minimum bend diameter. Eight times the cable outer diameter. (The short-term minimum bend diameter is to be used in all environmental and mechanical tests that specify a cable minimum bend diameter.)

Long-term minimum bend diameter. 16 times the cable outer diameter.

Minimum continuous length. The minimum continuous length of all cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, a conformance inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

Markings. Markings shall be in accordance with 3.9 of MIL-PRF-85045.

PERFORMANCE REQUIREMENTS:

Optical properties:

Maximum attenuation rate:

Temperature range 1. 3.5 dB per km at 850±20 nm, 1.0 dB per km at 1300±20 nm for class type MM fiber.

Temperature range 1. 0.4 dB per km at 1310±20 nm and 0.4 dB per km at 1550±20 nm for class type SM fiber.

Temperature range 2. 3.0 dB per km at 850±20 nm, 0.80 dB per km at 1300±20 nm for class type MM fiber.

Temperature range 2. 0.4 dB per km at 1310±20 nm and 0.3 dB per km at 1550±20 nm for class type SM fiber.

Point discontinuity (attenuation uniformity):

Method. The attenuation uniformity of each individual fiber shall be measured in accordance with TIA-455-78. The attenuation uniformity shall be measured at wavelengths of 850 nm and 1300 nm for multimode and at 1310 nm and 1550 nm for single mode.

Requirement. There shall be no discontinuities in attenuation along the length greater than 0.1 dB for single mode and no discontinuities in attenuation along the length greater than 0.2 dB for multimode.

Change in optical transmittance. For temperature range 1, measurements shall be made at 1300±20 nm for multimode and 1310±20 nm and 1550±20 nm for single mode. For temperature range 2, measurements shall be made at 850 nm±20 nm for multimode and 1550±20 nm for single mode.

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Measured fiber positions within the cable. Unless otherwise specified herein, the minimum number of active fibers required to be measured for a change in optical transmittance during mechanical and environmental testing shall be 12.

For shock and vibration testing only, four fibers shall be required to be monitored: two outer fibers and two inner fibers (as defined by the position in the array where the outer fibers are identified by positions 1, 2, 11, and 12 and the inner fibers by positions 5, 6, 7, and 8). For temperature range 1, shock and vibration testing only, single mode fiber, measurements at 1310±20 nm are required at a minimum (1550±20 nm is not required).

Crosstalk. Applicable.

Mechanical properties:

Low temperature flexibility. Applicable.

Cyclic flexing. For temperature range 1, the test shall be performed for 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. For temperature range 2, the test shall be performed for 500 cycles at 25±2 °C and 100 cycles at -55±2 °C. The minimum load shall be 1.0 kg. Changes in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Crush. Applicable except that the compression loading shall be not less than 250 newtons (N), shall be held for 1 minute and released, and shall be applied at a rate of not less than 250 N per minute.

Cable twist-bending. Applicable except 500 cycles shall be performed at 25±2 °C and 100 cycles at -28±2 °C for temperature range 1. For temperature range 2, the test shall be performed for 500 cycles at 25±2 °C and 100 cycles at -55±2 °C. The test load shall be 10 N. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Radial compression. Not applicable.

Impact (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2, except the test shall be performed for 50 cycles at 25±2 °C and for 20 cycles at the minimum operating temperature.

Altitude immersion (temperature range 2 only). The test shall be performed in accordance with TIA-455-15 using the setup and procedure specified in measurement 3305 of MIL-STD-1678-3. Instrument end connections shall not be submerged and shall be either routed outside the chamber or to an optical interface port. The change in optical transmittance shall be measured during and after the test. During each cycle, optical transmittance measurements shall be made at each ramp and during the plateau. The requirements for the change in optical transmittance shall be met. No damage to the cable shall occur, and there shall be no evidence of water penetration into the cable interior.

Flammability (60-degree angle). Applicable.

Tensile loading and elongation. Applicable, tensile loading ≥270 N.

Operating tensile loading. Applicable.

Cable element removability. Applicable with the following additions:

a. The end of the rolled flexible ribbon shall be easily formed by hand into a flat ribbon that fits into an interfacing rectangular trough (such as tool [fusion splice holder], housing [splice tray], and component [MT terminus/connector]).

b. The 12 colored fibers shall be visually inspected to verify that they are retained in the specified order of color once flattened and 20 cm of matrix material is removed from the end of the 12 fibers (except for any residual material left on the fibers from the matrix material shall be removed with a dry cotton cloth).

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c. The type of adhesive or matrix material used for the rolled flexible ribbon shall ensure fiber separation does not cause coating damage or leave residue.

Dynamic bend. Applicable. The tensile load shall be not less than 90 N.

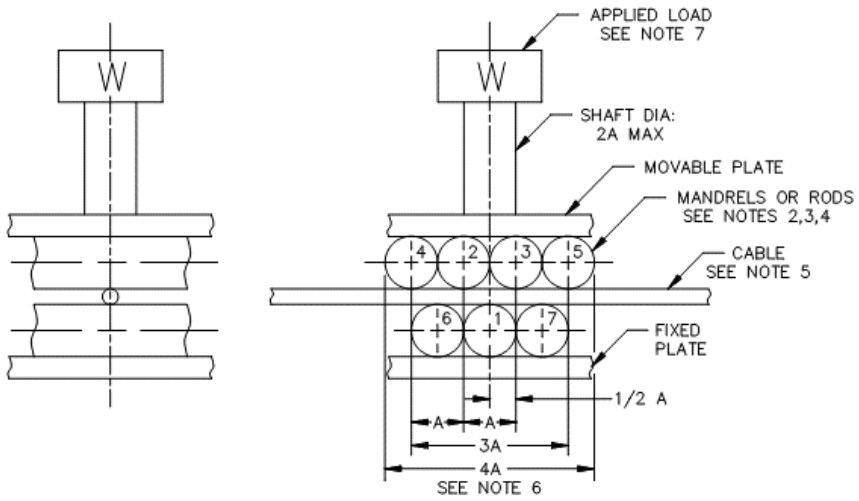
Corner bend. Applicable except that test force shall be 100 N (22 pounds [lbs]). The radius for the corner bend to use for single subunit cable shall be the same as specified for the short-term bend diameter.

Kink resistance (optical fiber cable component [OFCC] kink). Applicable.

Cable clamp (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2. The specimen length shall be two 1-meter minimum lengths using same specimens after performed kink, with 6 standard plastic straps spaced 100 mm (4 inches) apart, shall be held for 24 hours, and shall be measured for the change in optical transmittance. The change in optical transmittance shall not exceed the values specified in 3.5.2 of MIL-PRF-85045 during and after the test. Cable length shall be placed adjacent to the length of a 25-mm (approximately 1.0-inch) diameter metal mandrel. The plastic straps (adjustable, self-clinching, plastic tie down strap) used shall conform to MS3367-4 of SAE AS33671. Plastic straps shall be installed using a tool in accordance with SAE AS90387 with the tool maximum tension setting listed in SAE AS90387 for the strap part number being used. This configuration shall be specified for test purposes only as miniature nylon straps are not used on military aircraft. Visual inspection shall be performed after the test and shall reveal no kinking, cracking, splitting, tearing, or collapsing of the cable jacket.

Serpentine cable bend (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2. The specimens shall be tested between two plates on which mandrels (rods) with a 10-mm mandrel diameter (A) are welded or otherwise permanently affixed. A load shall be applied on the top plate of 50 N (11.2 lbs) for 24 hours, and the change in optical transmittance shall be measured. A change in optical transmittance shall not exceed the values specified in 3.5.2 of MIL-PRF-85045 during and after the test. A visual inspection shall be performed after the test and shall reveal no kinking, cracking, splitting, tearing, or collapsing of the cable jacket. The bottom plate shall have three mandrels placed adjacently and touching each other in the middle of the plate. The top plate shall have four mandrels placed adjacently and touching each other in the middle of the plate. Specimens shall be placed onto the bottom plate so that cable length is perpendicular to the mandrel's circumference. The top plate shall be placed onto the bottom plate for this test so that the inner two diameters of the mandrels on the top plate are placed between the two spaces of the three mandrels on the bottom plate. Applying and removal of the load shall not exceed a rate of 50 N per minute (11.2 lbs per minute). The fixture used to apply the loading shall conform to figure 2.

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NOTES:

1. Material (plates mandrels, shaft): metal.
2. Dimensions: A = mandrel diameter.
3. Mandrels 1, 2, and 3: primary applied loading members.
4. Mandrels 4, 5, 6, and 7: stability members.
5. Cable placed at centerline of applied load.
6. When the load is applied, the top mandrels shall fit into spacing between lower mandrels (as shown on the figure with centerlines). Guide pins or other means can be used to ensure this alignment.
7. Weight (W) is the applied load. Preference is to apply the load from a controlled device such as a tensile test machine or press.

FIGURE 2. Serpentine cable bend test fixture.

Hosing. Not applicable.

Dripping. Not applicable for temperature range 1. Applicable for temperature range 2 except that the test is performed at 200 °C (392 °F).

Water absorption. Not applicable.

Cable jacket tear strength. Applicable for temperature range 1, except the cable jacket tear strength shall be not less than 35 N per cm of jacket thickness for standard performance cable. Applicable for temperature range 2. Five strips of flat extruded jacket material shall be used.

Cable jacket mechanical strippability (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2 except that the test shall be performed on the outer cable jacket in accordance with TIA-455-178. Once the outer cable jacket is cut 360 degrees along the cable circumference (i.e., ring cut), the axial force to remove a 25-mm (1.0-inch) length of cable jacket shall not exceed 15.6 N (3.5 lbs). A strip rate of 250±50 mm per minute (10±2 inches per minute) shall be used.

Cable jacket material tensile strength and elongation. The tensile strength shall be 900 N per square centimeter (cm^2) minimum, elongation shall be 125 percent minimum. Restrictions on the setup and test procedure and compliance with data recording shall be in accordance with measurement 3410 of MIL-STD-1678-3.

Cable scraping resistance. Not applicable.

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Cable to cable abrasion (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2.

Cable shrinkage. Applicable for temperature range 1. Applicable for temperature range 2 except the total shrinkage of the 30 cm length shall be not greater than 3.18 mm (0.125 inch).

Durability of marking (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2.

Wicking (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2 except maximum water penetration shall be 25 mm (1 inch).

Cable strain relief pullout force. The pullout force of the cable strain relief (i.e., the cable member for strain relief to a connection device comprised of a material such as aramid yarn) shall be tested in accordance with TIA-455-6 by applying a 67-N (15-lb) load for a duration of 1 minute for a straight pull and a 33-N (7.5-lb) load for a duration of 1 minute at a 90-degree pull. One end on each length of the fiber optic cable shall be terminated onto the end of a MIL-PRF-29504 terminus (or other suitable fiber optic connector/capture method if approved by the qualifying activity prior to test sample fabrication).

a. Temperature range 1 only, splice tray. Specimen shall consist of being secured with fiberglass cloth tape to the cable entry end of a splice tray in accordance with MIL-STD-2042 method 2K1 and retaining adherence when the pullout force specified in MIL-DTL-24728/8 is exerted on the cable for both the straight pull and the 90-degree pull.

b. Temperature ranges 1 and 2, MT ferrule with a multi-fiber push on (MPO) type connector with a crimp barrel for a 2-mm crimp sleeve. Specimen shall consist of a commercial variant of the MPO connector with a 12-fiber MT ferrule. The connector shall be affixed to the 2-meter cable sample at one end and a suitable connector to mate with the optical instrumentation detector interface at the other cable sample end. A test cable shall interface the specimen through a commercial variant of the MT ferrule at one cable end and the optical source connection at the other cable end. The epoxy used to secure the cable strain relief to the terminus shall be in accordance with two-part liquid epoxy specified in appendix A of requirement 4101 of MIL-STD-1678-4 for temperature range 2 and as specified in appendix A of requirement 4102 of MIL-STD-1678-4 for temperature range 1.

An axial tensile load shall be applied on the detector end cable (specimen) through a mandrel or other approved means. The load shall not be applied to the coupling mechanism. For the MT ferrule (MPO) test configuration only, the change in optical transmittance shall be measured during and after the test and meet the optical requirement for this measurement. The cable strain relief shall be visually inspected after the test. No evidence of strain relief damage, breakage, or disengagement (pullout) shall occur.

Environmental properties:

Electromagnetic effects (temperature range 2 only). Not applicable for temperature range 1. The electromagnetic effects test shall be performed in accordance with measurement 3308 of MIL-STD-1678-3 for temperature range 2.

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Temperature cycling:

Method. The cable shall be tested in accordance with TIA-455-3 using the setup and procedure specified in measurement 3301 of MIL-STD-1678-3 with the temperature profile of "General applications" for temperature range 1 and the temperature profile of "Aircraft applications (-55 to 165 °C)" for temperature range 2. The cable shall be visually examined after the test. Test fixtures, if used, shall be of minimum mass and approved by the qualifying activity. No other mass shall be added inside the chamber. The change in optical transmittance shall be measured during and after the test. At a minimum for the measurements made during testing, an optical transmittance measurement shall be performed toward the end of each soak period (maintain step) after every cycle for temperature range 1 and after every five cycles for temperature range 2. The cable shall be visually examined after the test.

Temperature cycling with mandrel (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2. Testing shall be performed with a portion of cable wrapped around a 20±1-mm (0.79±0.04-inch) mandrel for five turns. The temperature profile for "General applications" of measurement 3301 of MIL-STD-1678-3 may be used in lieu of the table XIII of MIL-PRF-85045.

Temperature humidity cycling. Applicable except that the change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Storage temperature (temperature range 1 only). Applicable.

Temperature life (life aging). Applicable for temperature range 1. Applicable for temperature range 2 except it shall be performed as follows for the operating temperature range of -55 to 165 °C. The high exposure temperature shall be 165+5 °C, 0 °C. The specimens shall be tested in accordance with TIA/EIA-455-4 for the duration of 1,000 hours at the high exposure temperature. A visual inspection performed after the temperature life test for both temperature range 1 and temperature range 2 shall include inspection for the requirements specified under ribbon delamination and colorant.

Thermal shock. Applicable except for temperature range 2 perform at the operating temperature extremes.

Weathering. Not applicable.

Fluid immersion (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2 except that specimens and flat extruded jacket material strips shall be exposed to the fluids at the temperatures specified in MIL-STD-1678-3 measurement 3409, appendix A for temperature range 2. Specimens and flat extruded jacket material strips shall be maintained at ambient conditions for a minimum of 4 hours prior to fluid testing. After immersion, all specimens and flat extruded jacket material strips shall be removed, blotted, and suspended as specified in 4.7.6.9 of MIL-PRF-85045 prior to inspection. After testing, each specimen and flat extruded jacket material dumbbell shall be visually examined for conformance to the inspection requirements.

Vibration (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2. The specimens shall be tested for random vibration in accordance with measurement 3201 of MIL-STD-1678-3 and tested only for the second specified random vibration test under the test condition C synopsis. Optical discontinuities shall be measured during the test. A minimum of four specimens shall be monitored for optical discontinuity. The change in optical transmittance shall be monitored after the test. The specimens shall be visually examined after the test. The requirement for optical discontinuity shall be met during the test, and the requirement for the change in optical transmittance shall be met after the test.

Shock (temperature range 2 only). Not applicable for temperature range 1. Applicable for temperature range 2. The specimens shall be tested both in accordance with MIL-DTL-901, grade A, class I, implementing the restrictions and using the test fixtures specified in measurement 3202 of MIL-STD-1678-3 and in accordance with TIA-455-14, test condition D, implementing the further guidance and, as applicable, the fixtures as specified in measurement 3202 of MIL-STD-1678-3. Optical discontinuities shall be measured during the test. A minimum of four specimens shall be monitored for discontinuity. The change in optical transmittance shall be monitored after the test. The specimens shall be visually examined after the test. The requirement for optical discontinuity shall be met during the test, and the requirement for the change in optical transmittance shall be met after the test.

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Flame extinguishing and smoke generation. Not applicable.

Flaming smoke generation. The maximum specific optical density, D_m, shall be less than 225 when tested in the flaming condition in accordance with ASTM E662. The size and configuration of the test specimen shall conform to the 76- by 76-mm (3- by 3-inch) square specified in ASTM E662. The thickness of the test specimen shall be 2 mm (0.08 inch). The specimen shall be constructed by laying 76-mm (3-inch) lengths of OFCC adjacent to one another to form a 76-mm (3-inch) strip. Wire or cable clamps may be used to permanently hold the OFCC lengths in place. Longer OFCC lengths may be used if the top or bottom clamps are used to hold the OFCCs. If top or bottom clamps are used, the length of OFCC exposed outside of the clamp(s) shall be 76 mm (3 inches).

Continuous flame (temperature range 2 only). The radiant panel index shall be less than or equal to 20 when testing is performed in accordance with ASTM E162.

Paint susceptibility. Not applicable.

Chemical properties:

Acid gas generation. Applicable.

Halogen content (temperature range 1 only). <0.2 percent.

Toxicity index. Applicable.

Thermal vacuum outgassing (temperature range 2 only). Applicable for non-metallic materials only.

Qualification and conformance inspection. See table II.

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TABLE II. Qualification and conformance inspection.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/, 3/, 4/	Qualification temperature range 1	Qualification temperature range 2	Conformance inspection
I	Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	Three samples, 0.5 km each 5/	X	X	A
	Attenuation rate	3.5.1	4.7.4.1	Three samples, 0.5 km each 6/	X	X	A
	Point discontinuity	7/	7/	Three samples, 0.5 km each 6/	X	X	--
	Crosstalk	7/	7/	Three samples, 0.5 km each 6/	X	X	--
II	Electromagnetic effects	3.7.16	4.7.6.16	One sample, 6 m 8/		X	--
	Jacket red light transparency	7/	7/	One sample, 1 m 8/	X	X	--
III	Temperature cycling	7/	7/	Two samples, 0.5 km each 6/ (1 on reel, 1 off)	X	X	C
	Temperature cycling with mandrel	7/	7/	Two samples, 0.5 km each 9/ (1 on reel, 1 off)	--	X	--
	Temperature humidity cycling	3.7.3	4.7.6.3	Two samples, 0.5 km each 9/	X	X	C
	Storage temperature	3.7.4	4.7.6.4	Two samples, 0.5 km each 9/	X	--	--
	Low temperature flexibility (cold blend)	3.6.3	4.7.5.3	Two specimens, 8 m each 10/	X	X	B
	Cyclic flexing	3.6.4	4.7.5.4	Six specimens, 5 m each 11/ (two specimens at each temp)	X	X	--
	Crush	3.6.5	4.7.5.5	Two specimens, 5 m each 10/	X	X	--
	Cable twist bending	7/	7/	Six specimens, 5 m each 11/ (three specimens for each temperature)	X	X	C
	Impact	3.6.8	4.7.5.8	Two specimens, 5 m each 10/	--	X	C
	Altitude immersion	3.7.5	4.7.6.5	One specimen, 0.46 km 12/	--	X	--

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TABLE II. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/, 3/, 4/	Qualification temperature range 1	Qualification temperature range 2	Conformance inspection
III	Temperature life (life aging)	7/	7/	Two specimens, 300 m each 10/	X	X	C
	Tensile loading and elongation	3.6.1	4.7.5.1	Two specimens, 300 m each 13/	X	X	C
	Operating tensile loading	3.6.1.1	4.7.5.1.1	Two specimens, 300 m each 14/	X	X	C
	Freezing water immersion (ice crush)	3.7.7	4.7.6.7	Two specimens, 30 m each 10/	--	X	--
	Fungus resistance	3.8.4	4.8.4	Two specimens, 0.5 m each 10/	X	X	--
	Cable element removability	3.6.18	4.7.5.18	Two specimens, 0.5 m each 10/	X	X	C
	Flammability (60-degree angle)	3.7.12.1	4.7.6.12.1	Two specimens, 6 m each 10/	X	X	C
IV	Thermal shock	3.7.2	4.7.6.2	One specimen, 0.49 km each 6/ (on reel)	X	X	--
	Fluid immersion	7/	7/	One specimen, 1 m 15/ and 3 flat extruded jacket material strips 16/ for each specified fluid	--	X	C
	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	One specimen, 30 m 15/	X	X	--
	Vibration	7/	7/	Four specimens, 30 m each 15/	--	X	--
	Shock	7/	7/	Four specimens, 30 m each 15/	--	X	--
	Dynamic bend	3.6.2	4.7.5.2	One specimen, 150 m 17/	X	X	B
	Corner bend	3.6.9	4.7.5.9	Two specimens, 5 m each 18/	X	X	--
	Kink resistance	7/	7/	Two specimens, 1 m each 18/	X	X	--
	Cable clamp	7/	7/	Two specimens, 1 m each 18/	--	X	--

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TABLE II. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/, 3/, 4/	Qualification temperature range 1	Qualification temperature range 2	Conformance inspection
IV	Serpentine cable bend	7/	7/	Two specimens, 1 m each 18/	--	X	--
V	Dripping	3.6.13	4.7.5.13	One sample, 30 cm 15/	--	X	--
	Cable jacket tear strength	3.6.14	4.7.5.14	Five flat extruded jacket material strips 19/	X	X	C
	Cable jacket mechanical strippability	7/	7/	One specimen 18/, 20/	--	X	--
	Cable jacket material tensile strength and elongation	3.6.15	4.7.5.15	Five specimens 16/	X	X	--
	Cable abrasion resistance, cable-to-cable	3.6.16	4.7.5.16.2	Two specimens, 2 m each 17/	--	X	C
	Cable shrinkage	3.6.17	4.7.5.17	Three specimens, 0.5 m 18/	X	X	C
	Durability of identification marking	3.6.19	4.7.5.19	Three specimens, 2 m each 18/	--	X	--
	Flaming smoke generation	7/	7/	One specimen, 3 m 21/, 22/	X	X	C
	Continuous flame	7/	7/	One specimen, 22/	--	X	C
	Wicking	3.7.10	4.7.6.10	Two specimens, 2 m each 18/	--	X	C
	Cable strain relief pullout force	7/	7/	Five specimens, 2 m each 18/	--	X	--
VI	Acid gas generation	3.8.1	4.8.1	One specimen 1 m 22/	X	X	C
	Halogen content	3.8.2	4.8.2	One specimen, 1 m 22/	X	--	C
	Toxicity index	3.8.3	4.8.3	One specimen, 1 m 22/	X	X	C
	Thermal vacuum outgassing	3.3.6.1	4.8.5.1	One specimen, 1 m 22/	--	X	C

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TABLE II. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/, 3/, 4/	Qualification temperature range 1	Qualification temperature range 2	Conformance inspection
VII	Ribbon strippability	7/	7/	Three specimens, 2 m each 22/	X	X	--
	Ribbon twist	7/	7/	Three specimens, 1 m each 22/	X	X	--
	Ribbon delamination	7/	7/	Three specimens, 2 m each 22/	X	X	--
	Ribbon separation	7/	7/	One specimen, 0.5 m 22/	X	X	--

NOTES:

1/ Tolerance on 0.5-km length shall be ±5 percent, provided that results are normalized to 1 km.

2/ Tolerance on shorter lengths shall be ±5 percent.

3/ A sample shall be the initial 0.5-km cable length.

4/ A specimen shall be a cable length that is cut from the initial 0.5-km cable length (i.e., the sample) or is cut from a longer previous cut length (i.e., cut from a longer specimen).

5/ The visual and mechanical inspection shall only be conducted on a 2-meter section of each sample.

6/ The same samples as used in the visual and mechanical inspection shall be used.

7/ As stated under the applicable tests in this specification sheet.

8/ A specimen cut from one of the visual and mechanical inspection test samples shall be used.

9/ The same samples as used in the temperature cycling test shall be used.

10/ A specimen cut from each sample used in the temperature cycling test shall be used.

11/ Three specimens cut from each sample used in the temperature cycling test shall be used.

12/ One of the samples used in the temperature cycling test shall be used.

13/ A specimen cut from each specimen used in the temperature life test shall be used.

14/ The same specimen as used in the tensile loading and elongation test shall be used.

15/ A specimen cut from the specimen used in the thermal shock test shall be used.

16/ Flat extruded jacket material as specified in 4.7.5.15 of MIL-PRF-85045 shall be used.

17/ A specimen cut from the specimen used in the thermal shock test or one of the specimens used in the tensile loading and elongation test shall be used.

18/ Specimens cut from the specimen used in the thermal shock test shall be used.

19/ Flat extruded jacket material strips (i.e., strips of flat extruded material with the same composition of the cable jacket and of sufficient dimensions in which dumbbells can be cut) prepared as specified in 4.7.5.14 cited method and obtained from flat extruded material that underwent the thermal shock test shall be used.

20/ The sufficient length to obtain five stripping trials shall be used.

21/ Use a 76-mm (3-inch) length of cable adjacent to one another to form a 76-mm (3-inch) strip. For a 2-mm (0.08-inch) diameter cable, an approximate length of 3 meters shall be used.

22/ A specimen cut from one of the samples used in the temperature cycling test or specimen used in the thermal shock test shall be used.

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Qualification and conformance inspection by similarity:

Similarity for temperature range 1 single mode and multimode. Manufacturers who are qualified under this specification sheet for the temperature range 1 rolled flexible ribbon cable with single mode fiber (M85045/33-02) and whose temperature range 1 rolled flexible ribbon cable with multimode fiber passes the tests and inspections specified in table III as specified herein shall be qualified under this specification sheet for the temperature range 1 rolled flexible ribbon cable with multimode fiber (M85045/33-01). This qualification by similarity shall be applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber has been changed from single mode optical fiber to multimode optical fiber. Testing may be performed on a single length of cable with a minimum length of 0.5 km.

TABLE III. Qualification and conformance by similarity for M85045/33-01 (cable when qualified to M85045/33-02 cable).

Group	Qualification inspection	Qualification inspection M85045/33-01 1/, 2/	Conformance inspection M85045/33-01 1/, 2/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Temperature cycling	X	C
	Low temperature flexibility (cold bend)	X	B
IV	Thermal shock	X	--
	Kink resistance	X	--
V	Flaming smoke generation	X	C

NOTES:

1/ Where manufacturers are qualified for M85045/33-02 and are pursing qualification for M85045/33-01.

2/ Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order shall be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

Similarity for temperature range 2 single mode and multimode. Manufacturers who are qualified under this specification sheet for the temperature range 2 rolled flexible ribbon cable with multimode fiber (M85045/33-03) and whose temperature range 2 rolled flexible ribbon cable with single mode fiber passes the tests and inspections specified in table IV as specified herein, are qualified under this specification sheet for the temperature range 2 rolled flexible ribbon cable single mode fiber (M85045/33-04). This qualification by similarity shall be applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber has been changed from multimode optical fiber to single mode optical fiber. Testing may be performed on a single length of cable with a minimum length of 0.5 km.

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TABLE IV. Qualification and conformance by similarity for M85045/33-04 cable when qualified to M85045/33-03 cable.

Group	Qualification inspection	Qualification inspection M85045/33-04 1/, 2/	Conformance inspection M85045/33-04 1/, 2/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Temperature cycling	X	C
	Temperature humidity cycling	X	C
	Cyclic flexing	X	--
IV	Thermal shock	X	--

NOTES:

1/ Where manufacturers are qualified for M85045/33-03 and are pursing qualification for M85045/33-04.

2/ Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. The test order shall be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

Similarity for temperature range 1 cable with multiple subunit cables. Manufacturers who produce products for M85045/34 or M85045/35 and this specification sheet for the temperature range 1 rolled flexible ribbon cable and whose product to this specification sheet passes the tests and inspections specified in table V are qualified under this specification sheet for the temperature range 1 rolled flexible ribbon cable. This qualification by similarity shall be applicable if the product produced to this specification sheet was used in the production of the already qualified product under M85045/34 or M85045/35. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

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TABLE V. Qualification and conformance by similarity for M85045/33 temperature range 1 cable when qualified to M85045/34 or M85045/35.

Group	Qualification inspection	Qualification inspection M85045/33-01 1/, 2/	Conformance inspection M85045/33-01 1/, 2/	Qualification inspection M85045/33-02 2/, 3/	Conformance inspection M85045/33-02 2/, 3/
I	Visual and mechanical	X	A	X	A
	Attenuation rate	X	A	X	A
III	Temperature cycling	X	C	X	C
	Low temperature flexibility (cold bend)	X	B	X	B
IV	Thermal shock	X	--	X	--
	Kink resistance	X	--	X	--
V	Flaming smoke generation	X 4/	C 4/	X 4/	C 4/

NOTES:

1/ Where manufacturers are qualified for M85045/34-01P or M85045/35-01P and are pursing qualification for M85045/33-01.

2/ Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order shall be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

3/ Where manufacturers are qualified for M85045/34-02P or M85045/35-02P and is pursing qualification for M85045/33-02.

4/ This qualification or conformance test shall be required for only one of the PINs in this specification sheet under the provision that the same cable jacket compound (including color) is used for both M85045/33-01 and M85045/33-02.

Intended use. This cable is intended for use inside of protected enclosures (such as convoluted tubing or cable wrap) and cable harnesses and is not intended for installation in the overheads or cableways outside protective enclosures. Examples of use include jumper cordage and pigtails terminated in 12-fiber MT ferrules (and inserted into both backplane and circular connectors). Also, this cable is intended for use as a core component where multiple subunit cables are housed within an outer jacket (i.e., a multiple rolled flexible ribbon fiber optic cable).

Application restriction. A cable produced under this specification sheet with the 50-/125-micron optical fiber (i.e., M85045/33-03) shall not be used for Navy shipboard applications.

Bandwidth measurements (multimode cables only). For this cable construction, the bandwidth is measured as part of the requirements of the optical fiber to be cabled. The bandwidth value does not change once the optical fiber is cabled. Since the bandwidth is provided as one of the cable-measured parameters, the acceptable bandwidth for the optical fiber used in this cable (and by extension, the cable) is provided for informational purposes.

For cable with MM 62.5-/125-micron fiber. ≥200 megahertz (MHz)-km at 850±20 nm, ≥500 MHz-km at 1300±20 nm for cable with MM fiber.

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For cable with MM 50-/125-micron fiber. ≥1,500 MHz-km at 850±20 nm, ≥500 MHz-km at 1300±20 nm for cable with MM fiber.

Restricted bandwidth for cable with MM 50-/125-micron fiber: ≥4,700 MHz-km at 850±20 nm.

DEFINITIONS:

Rolled. The term rolled is used to emphasize the requirement for the ribbon to be rolled longitudinally within the cable to increase the fiber packing density (allow for a specified number of fibers for the given cable diameter).

Flexible. The term flexible is used to emphasize the requirement that the application of the adhesive matrix is applied intermittently to allow the rolled flexible ribbon to take a planar shape for connector termination and splicing and take a circular shape when rolled longitudinally within the cable.

Referenced documents. In addition to MIL-PRF-85045, this document references the following:

MIL-DTL-901
MIL-DTL-24728/8
MIL-PRF-29504
MIL-PRF-49291/6
MIL-PRF-49291/7
MIL-PRF-49291/11
MIL-PRF-49291/12
MIL-STD-1678-1
MIL-STD-1678-3
MIL-STD-1678-4
MIL-STD-2042
ASTM E162
ASTM E662
SAE AS33671
SAE AS90387
TIA-455-3
TIA/EIA-455-4
TIA-455-6
TIA-455-14
TIA-455-15
TIA-455-78
TIA-455-141
TIA-455-178
TIA-598

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CONCLUDING MATERIAL

Custodians:
- Army – CR
- Navy – SH
- Air Force – 85
- NASA – NA

Review activities:
- Army – AR, AV, MI
- Navy – AS, EC, OS, YD
- Air Force – 02, 13, 19, 70, 93
- DLA – CC
- MISC – MDA

Civil agency:
- GSA – FAS

Preparing activity:
- Navy – SH

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